JobLens

Job Market Insights & Salary Prediction System

Data Engineering + Data Science / ML

2,536	$35.7k	0.4706	5
Cleaned Roles	MAE	R^2 Score	Clusters

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GitHub: github.com/Sujoy-004/JobLens
Live Demo: joblens-capstone-kiit0001.vercel.app

1. Problem Statement

The data science and technology job market changes rapidly, and salary varies significantly across role families, seniority levels, company size, and geography. Job seekers and reviewers need a compact, accessible way to benchmark compensation and understand job archetypes without relying on paid or region-locked tools.

JobLens addresses that gap with a reproducible data pipeline that cleans the public salary dataset, extracts skill signals from job titles, trains prediction and clustering models, and publishes the results through a static Vercel dashboard — accessible to anyone, with no backend dependency or subscription required.

2. Solution / Features

The project is split into three clearly defined layers so the notebook, models, and frontend remain easy to explain, validate, and reproduce from the exported files.

Component	What it covers	Output
Data Engineering	Load, clean, and standardise ds_salaries.csv; remove duplicates and outliers; derive skill flags from job titles.	cleaned_data.csv
Machine Learning	Train a Gradient Boosting Regressor for salary prediction and K-Means for five job groups.	predictions.csv and clusters.csv
Static Frontend	Read dashboard_data.json and render the predictor, charts, and cluster table in the browser.	Vercel-ready dashboard

Pipeline Architecture

Stage	What it does	Output
Data Ingestion	Load ds_salaries.csv, validate schema, inspect dtypes	Raw DataFrame
Data Cleaning	Drop duplicates, remove nulls, trim text, remove salary outliers	Cleaned DataFrame
Feature Engineering	Create 7 skill flags, encode categoricals, normalise rare titles	Feature matrix
Regression	GradientBoostingRegressor with 80/20 split	predictions.csv
Clustering	StandardScaler → KMeans(k=5) → PCA(2D)	clusters.csv
JSON Export	Aggregate charts, table data, and scatter sample	dashboard_data.json
Frontend	Fetch JSON and render the dashboard	Interactive web app

3. Screenshots

The following captures are taken directly from the deployed frontend. They demonstrate the predictor interface, salary analytics dashboard, PCA cluster scatter plot, and the skill-based job group table.

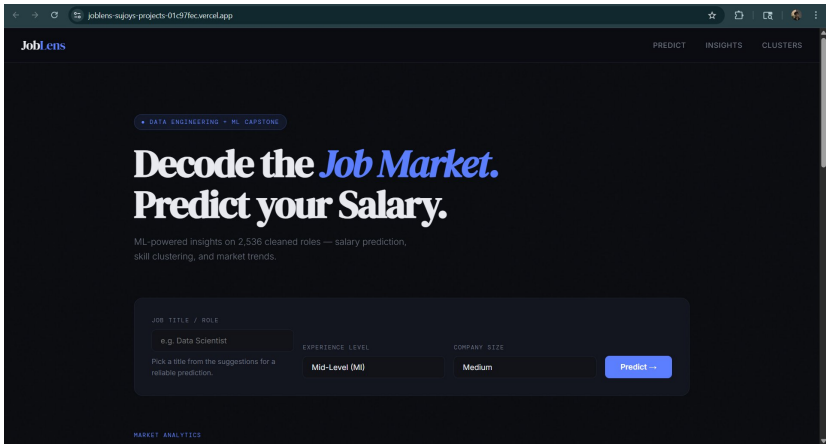

Fig 1: Homepage — Salary Predictor input form with job title, experience level, and company size selectors.

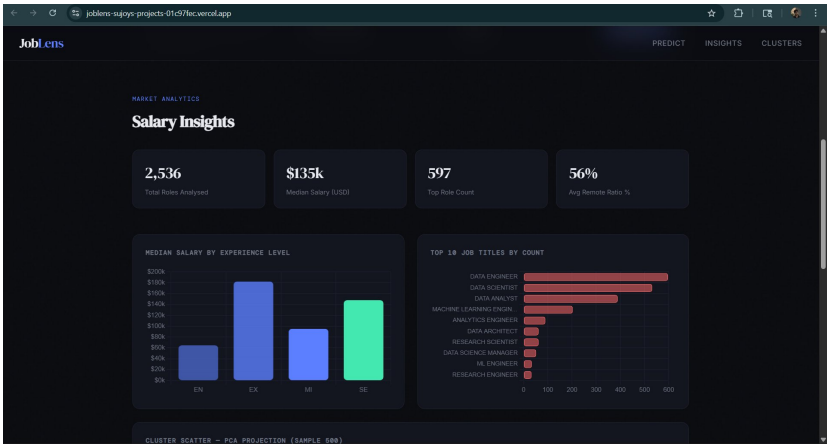

Fig 2: Market Analytics dashboard — key stats (2,536 roles, $135k median), salary-by-experience bar chart, and top-10 job titles.

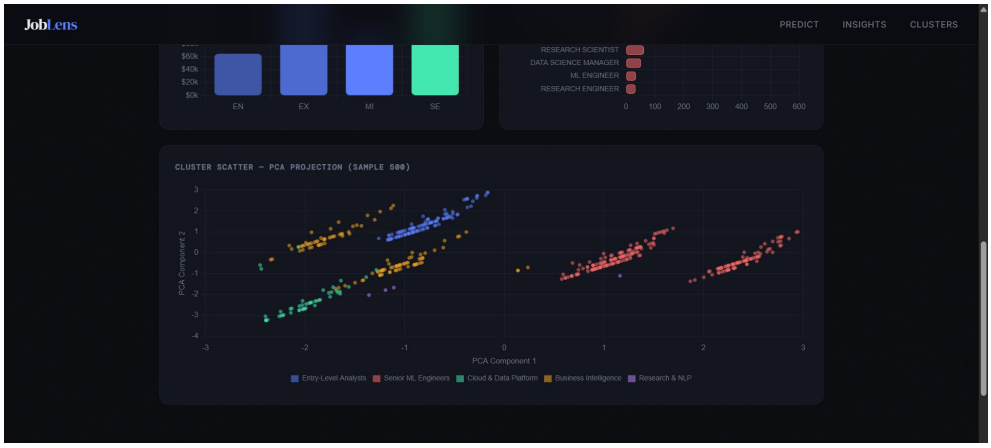

Fig 3: PCA Cluster Scatter Plot — 500 sampled roles projected to 2D, colour-coded by the five K-Means clusters.

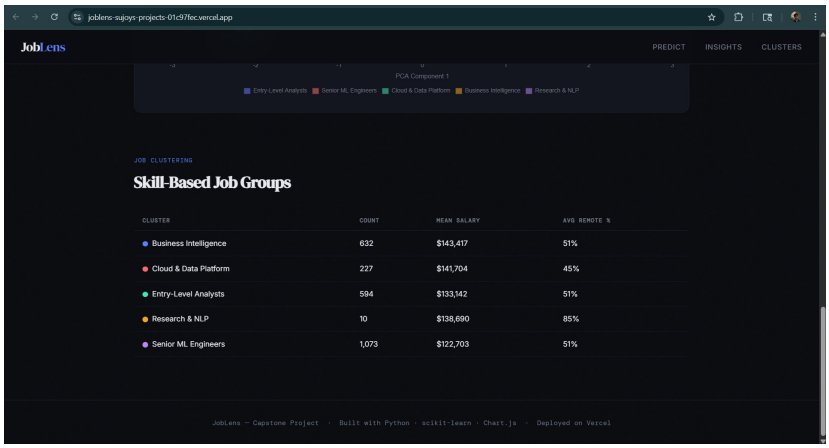

Fig 4: Skill-Based Job Groups table — cluster count, mean salary, and average remote ratio per group.

4. Tech Stack

The stack is intentionally lightweight to ensure the project is easy to deploy, explain during evaluation, and reproduce from the notebook and exported files.

Layer	Technology	Why it fits
Data Processing	Python, Pandas, NumPy	Reliable cleaning and transformation of tabular data
Modelling	scikit-learn — GradientBoostingRegressor and K-Means	Strong tabular baseline and readable clusters
Visualisation	PCA and Chart.js	Compact 2D cluster plots and interactive browser charts
Deployment	Static HTML, CSS, JS on Vercel	No backend, zero hosting friction, instant CDN delivery

Model performance (safe to report in evaluation): MAE ≈ $35,700 and $R^2 \approx 0.47$. The clustering model uses five interpretable groups: Entry-Level Analysts, Senior ML Engineers, Cloud & Data Platform, Business Intelligence, and Research & NLP.

5. Unique Points

- **Skill Inference Without NLP Libraries**

  Keyword-based tags are derived directly from job titles with no spaCy or API dependency, keeping the pipeline lightweight and fully offline.

- **Zero-Backend Architecture**

  A single static JSON file powers the entire dashboard. There is no server, no database, and no runtime cost — just a Vercel CDN edge.

- **Graceful Demo Fallback**

Demo data keeps the application fully functional and explorable even if the JSON file is missing or the network is unavailable.

- **Dual-Model Pipeline**
  Regression and clustering share one engineered DataFrame, so feature work is done once and both models benefit simultaneously.

- **PCA-to-Frontend Flow**
  Cluster coordinates are computed once in the notebook and embedded directly into dashboard_data.json, rendered in Chart.js with no extra computation at runtime.

6. Future Improvements

The current submission is complete and functional. The following improvements are planned for the next iteration to improve inference freshness, model interpretability, and market coverage.

Priority	Improvement	Impact
High	Add a FastAPI inference endpoint	Live model calls instead of a pre-computed lookup table
High	Use LLM-based skill extraction	Deeper title understanding beyond keyword matching
Medium	Expand the dataset with more recent records	Lower MAE and improve cluster granularity
Medium	Add SHAP explanations	Clearer, feature-level prediction explanations for end users
Medium	Add time-series salary trends	Show salary movement over time across roles and regions
Low	Role comparison view	Side-by-side salary and skill comparison between two roles

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This document is aligned with the live dashboard, the exported model outputs, and the static Vercel deployment. The submission package includes the project ZIP, GitHub repository, and this documentation PDF as required.